

Logistic Regression

↳ predictive technique used for classification problems.

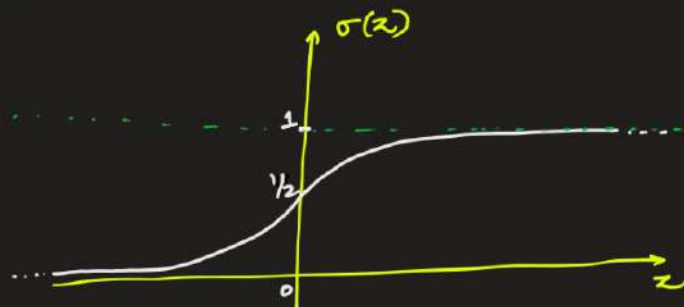
Unlike linear regression which outputs continuous numbers, logistic regression transforms its output using the logistic sigmoid function to return a probability value which can be mapped to two or more discrete classes.

Linear Regression:- Predicts a continuous outcome based on the sum of the weighted inputs. It fits a line to data points

Logistic Regression:- Predicts a categorical outcome. Instead of fitting a line, it fits an "S-shaped" logistic function, known as the sigmoid function. The output of the sigmoid function is used to categorize the data into classes (usually 0/1)

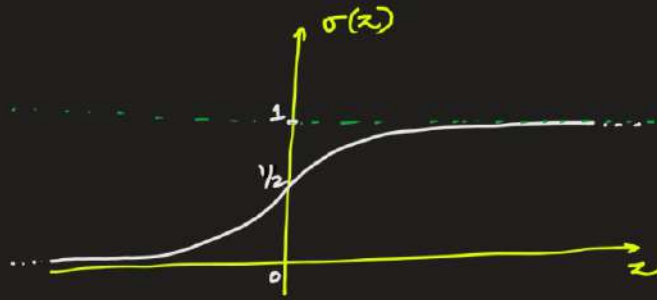
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid
function



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

↑
Sigmoid
function



- The output is bounded between 0 and 1, making it suitable for modeling a probability

Mathematical Formulation of Logistic Regression

Odds: In probability, is the ratio of the probability that the event will occur to the probability that it will not occur

$$\text{Odds} = \frac{P(\text{event})}{1 - P(\text{event})}$$

Log odds: The log odds is simply the natural logarithm ($\log_{\text{base } e}$) of the odds

$$\text{Log odds} = \log_e \left(\frac{P(\text{event})}{1 - P(\text{event})} \right)$$

Model Formulation: Logistic regression models the log odds of the dependent binary outcome as a linear relationship with the independent variables (predictors)

$$\log \left(\frac{P(Y=1|X)}{1-P(Y=1|X)} \right) = \beta_0 + \beta_1 x_1 + \beta_2 \underline{x_2} + \dots + \beta_p x_p$$

Each unit increase in x_k multiplies the odds of the outcome by e^{β_k}

Suppose that the model is predicting the likelihood of a disease based on age, the coefficient of age is $\beta_{\text{age}} = 0.05$.

↳ For every additional year of age, the odds of having a disease are multiplied by $e^{0.05} \approx 1.05$, indicating a 5% increase in the odds each year.

$$\rightarrow \log \left(\frac{P(Y=1|X)}{1-P(Y=1|X)} \right) = \cancel{\beta_0 + \beta_1 x_1 + \dots} + \beta_{\text{age}} \overset{\uparrow}{x_{\text{age}}} + \dots$$

$$\text{odds} \propto e^{\beta_{\text{age}} \cdot x_{\text{age}}} = e^{0.05 \times x_{\text{age}}}$$

$$e^{0.05} \rightarrow e^{0.05 \times 2} \rightarrow e^{0.05 \times 3}$$

$$\text{multiplying } e^{0.05} \approx 1.05$$

$$\begin{aligned} \log \left(\frac{P(Y=1|X)}{1-P(Y=1|X)} \right) &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p \\ &= \vec{X} \overset{\rightarrow}{\beta} \\ &\quad \underset{1 \times (p+1)}{\quad} \quad \underset{(p+1) \times 1}{\quad} \\ &= \underset{1 \times 1}{X} \end{aligned}$$

$$\begin{aligned}
 \therefore \log\left(\frac{p}{1-p}\right) &= z & \Rightarrow \frac{p}{1-p} &= e^z \\
 & & \Rightarrow p &= e^z - e^z p \\
 & & \Rightarrow (1+e^z)p &= e^z \\
 & \Rightarrow p &= \frac{e^z}{1+e^z} &= \frac{e^{x\beta}}{1+e^{x\beta}} \\
 & & &= \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-x\beta}} \\
 & & &= \sigma(z) = \sigma(x\beta) \\
 & & &= \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p)
 \end{aligned}$$

$$P(Y=1 | X=x_1, x_2, \dots, x_p) = \sigma(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)$$

$$\therefore P(Y=0 | X=x_1, x_2, \dots, x_p) = 1 - \sigma(z)$$

Maximum Likelihood Estimation for Logistic Regression

$$\begin{aligned}
 P(Y=1|X) &= \sigma(z) \\
 P(Y=0|X) &= 1 - \sigma(z)
 \end{aligned}
 \left. \begin{array}{l} \beta_0, \beta_1, \dots, \beta_p \\ \text{Bernoulli} \\ \text{distribution} \end{array} \right\}$$

objective:- Find the optimal values of $\vec{\beta}$ given a labelled training dataset on n data points $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

	X				$Y (0/1)$
$\rightarrow x_1$	x_{11}	x_{21}	$\dots$	x_{p1}	0
x_2	x_{12}	x_{22}	$\dots$	x_{p2}	1
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
x_N	x_{1N}	x_{2N}	$\dots$	x_{pN}	0

$$P(Y=1|X) = \sigma(z)$$

what β maximizes the likelihood of the sample?

For $y=1$, $p(x; \beta) = \underline{p(x)}$

For $y=0$, $p(x; \beta) = \underline{1-p(x)}$

$$L(x_i; \beta) = p(x_i)^{y_i} \cdot (1-p(x_i))^{1-y_i}$$

↓
we have n such independent points

$$L(x; \beta) = \prod_{i=1}^N p(x_i)^{y_i} \cdot (1-p(x_i))^{1-y_i}$$

$$\log L(x; \beta) = \sum_{i=1}^N \left[y_i \log p(x_i) + (1-y_i) \log (1-p(x_i)) \right]$$

$$= \sum_{i=1}^N \log (1-p(x_i)) + \sum_{i=1}^N y_i \left[\log \frac{p(x_i)}{1-p(x_i)} \right] \rightarrow \text{log odds} = z_i = x_i \beta$$

$$= \left[\sum_{i=1}^N \left[-\log (1 + e^{x_i \beta}) \right] + \sum_{i=1}^N y_i x_i \beta \right]$$

$$= - \sum_{i=1}^N \log (1 + e^{\beta_0 + x_{i1} \beta_1 + x_{i2} \beta_2 + \dots + x_{ip} \beta_p}) + \sum_{i=1}^N y_i (\beta_0 + x_{i1} \beta_1 + \dots + x_{ip} \beta_p)$$

$$\frac{\partial \log L(x; \beta)}{\partial \beta_0} \stackrel{=0}{=} \dots$$

↗

$$\frac{\partial \log L(x; \beta)}{\partial \beta_1} \stackrel{=0}{=} \dots$$

↗

$$\frac{\partial \log L(x; \beta)}{\partial \beta_p} \stackrel{=0}{=} \dots$$

↗

The derivatives of the log-likelihood function do NOT simplify to a form that can be easily solved for $\vec{\beta}$. The presence of $\sigma(x\beta)$, which includes both $e^{x\beta}$ and its reciprocal, results in transcendental equations that typically don't have closed-form solutions.

We use Numerical optimization techniques like Gradient Ascent or Newton-Raphson method to find the optimal values of β .

$$\log L(x; \beta) = - \sum_{i=1}^N \log(1 + e^{B_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ip}\beta_p}) + \sum_{i=1}^N y_i (B_0 + x_{i1}\beta_1 + \dots + x_{ip}\beta_p)$$

$$\frac{\partial \log L(x; \beta)}{\partial \beta_0} = - \sum_{i=1}^N \frac{e^{x_i \beta}}{1 + e^{x_i \beta}} + \sum_{i=1}^N y_i = - \sum_{i=1}^N \sigma(z_i) + \sum_{i=1}^N y_i$$

$$\frac{\partial \log L(x; \beta)}{\partial \beta_k} \quad k \neq 0 = - \sum_{i=1}^N \frac{e^{x_i \beta}}{1 + e^{x_i \beta}} \cdot x_{ki} + \sum_{i=1}^N y_i \cdot x_{ki} = - \sum_{i=1}^N \sigma(z_i) \cdot x_{ki} + \sum_{i=1}^N y_i \cdot x_{ki}$$

$$\boxed{\frac{\partial \log \text{likelihood}}{\partial \beta_0} = \sum_{i=1}^N (y_i - \sigma(z_i))}$$

$$\boxed{\frac{\partial \log \text{likelihood}}{\partial \beta_k} = \sum_{i=1}^N (y_i - \sigma(z_i)) \cdot x_{ki}}$$

Gradient ascent to find $\vec{\beta}$ values

① Initialize the parameters $\vec{\beta}$ ($\beta_0, \beta_1, \dots, \beta_P$)

↳ Zero initialization

↳ Random

② Calculate $\frac{\partial \log \text{likelihood}}{\partial \beta_k}$; $0 \leq k \leq P$

③ update β values in the direction of gradient

$$\beta_k = \beta_k + \alpha \cdot \left(\frac{\partial L}{\partial \beta_k} \right) \rightarrow \text{using eqn ②}$$

④ Stop at convergence

For prediction of class of a new point;

$$\log \left(\frac{P(y=1|x=\text{new point})}{1 - P(y=1|x=\text{new point})} \right) = z = \underset{\substack{\uparrow \\ \text{new point}}}{\vec{x}} \vec{\beta}$$

$$\begin{aligned} \hookrightarrow P(y=1|x=\text{new point}) &= \sigma(z) = \sigma(x\beta) \\ &= \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-\underset{\substack{\uparrow \\ \text{new point}}}{x\beta}}} \end{aligned}$$

$$P(y=0|x=np) = 1 - P(y=1|x=np)$$

eg.

Person	Income (x)	Purchase? (y) (0/1)
→ 1	40	0
→ 2	55	1
→ 3	30	0
→ 4	70	1

$x = 65$, predict they will purchase or not using logistic regression.

Sol. $x = \beta_0 + \beta_1 x$ $\sigma(z) = \frac{1}{1+e^{-z}}$

$$\frac{\partial \text{loglik}}{\partial \beta_0} = \sum_{i=1}^N (y_i - \sigma(z_i)) \quad , \quad \frac{\partial \text{loglik}}{\partial \beta_1} = \sum_{i=1}^N (y_i - \sigma(z_i)) \cdot x_i$$

Initialization $\beta_0 = 0, \beta_1 = 0, LR, \alpha = 0.01$

1st iteration:- $\frac{\partial L}{\partial \beta_0} = (0 - \frac{1}{1+e^{-0}}) + (1 - \frac{1}{1+e^0}) + (0 - \frac{1}{1+e^0}) + (1 - \frac{1}{1+e^0}) = -\frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = 0$

$$\frac{\partial L}{\partial \beta_1} = (-\frac{1}{2}) \cdot 40 + (\frac{1}{2}) \cdot 55 + (-\frac{1}{2}) \cdot 30 + (\frac{1}{2}) \cdot 70 = 27.5$$

$$\beta_0 = \beta_0 + \alpha \cdot \frac{\partial L}{\partial \beta_0} = 0 + 0.01 \times 0 = 0$$

$$\beta_1 = \beta_1 + \alpha \cdot \frac{\partial L}{\partial \beta_1} = 0 + 0.01 \times 27.5 = 0.275$$

2nd iteration:- $\frac{\partial L}{\partial \beta_0} = (0 - \frac{1}{1+e^{(0+0.275 \times 40)}}) + \dots$

$$\frac{\partial L}{\partial \beta_1} = \dots$$

$\boxed{\beta_0}, \boxed{\beta_1}$

$$p(y=1 | x=65) = \sigma(z) = \frac{1}{1+e^{-(\beta_0 + \beta_1 \times 65)}} \dots$$